llmXive Automated Review of minWM: A Full-Stack Open-Source Framework for Real-Time Interactive Video World Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the fidelity of citations to the attributed evidence.

Citation and Equivalence Claims: In Section 3.2, under "Stage 2 (option b): causal CD initialization," the authors state that Causal Forcing++ replaces ODE distillation with causal consistency distillation (CD), claiming the two are "theoretically equivalent" and citing `zhao2026causal`. While consistency distillation is often derived as a limit or alternative formulation of ODE distillation, strict theoretical equivalence usually requires specific conditions (e.g., the teacher model being a perfect ODE solver or specific noise schedules). Without these caveats, the claim of absolute equivalence is slightly overstated. The text should qualify this as "equivalent under standard consistency training assumptions" or similar to maintain scientific precision.

Terminology Precision: In Section 4.2, the text states the model achieves a "$223.75\times$ first-frame latency reduction." Mathematically, a "reduction" usually refers to the difference ($L_{old} - L_{new}$) or the percentage decrease. The value 223.75 is the *speedup factor* (ratio L_{old}/L_{new}). While the table correctly labels the column "Speedup," the prose conflates the ratio with the concept of reduction. This is a minor semantic inaccuracy that should be corrected to "speedup of" or "latency reduction of 99.5%."

Attribution of Methodology: The paper relies heavily on `li2026cameras` (PRoPE) for the camera injection mechanism described in Section 3.1. The mathematical formulation provided (block-diagonal transformation D_t^{PRoPE} injected into self-attention via GTA form) is highly specific. Given that `li2026cameras` is a future-dated citation (2026), it is critical to ensure that this specific architectural modification is indeed the core contribution of that work and not a novel combination by the current authors. If PRoPE is a standard method in that preprint, the citation is accurate; if the authors modified the standard PRoPE injection, the claim that they "adopt PRoPE" without noting modifications is misleading.

Data Claims: The ablation studies (Section 4.3) make qualitative claims about "uncontrollable" vs. "strong controllability" based on visual inspection of figures (e.g., Fig. 4, 5, 6). Since the figures are not rendered in the text, the claim that the model is "completely uncontrollable" at 1-2k steps is an interpretation of visual data. This is acceptable as a qualitative report, provided the authors do not claim quantitative metrics (like a specific controllability score) that are not present in the text or tables. The current text avoids specific numbers for these qualitative claims, which is appropriate.

Overall, the claims are largely supported, but the "theoretical equivalence" and "latency reduction" phrasing require minor tightening to be rigorously accurate.

Action items.

- **[writing]** In Section 3.2 (Stage 2 option b), the paper claims causal CD is 'theoretically equivalent' to causal ODE distillation citing Zhao et al. (2026). This equivalence typically holds only under specific assumptions (e.g., linear ODEs or specific noise schedules) which are not stated. The claim should be qualified to avoid implying universal equivalence.
- **[writing]** Table 1 reports a 223.75x speedup for HY1.5. The calculation (771.041 / 3.446) is correct, but the text claims this is a 'reduction' in latency. While colloquially acceptable, 'speedup factor' is the precise term for the ratio, whereas 'reduction' implies a subtraction (e.g., 99.5% reduction). Clarify terminology to ensure precision.
- **[writing]** The paper cites 'li2026cameras' (PRoPE) for camera injection. As this is a 2026 citation, verify that the method described (GTA form injection with block-diagonal projective matrices) is accurately attributed to that specific preprint and not a conflation with other camera-conditioning methods (e.g., ControlNet-style or standard cross-attention).

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure is a complete mismatch for its caption; it shows a fantasy scene with game controls instead of the required experimental results on batch size training effects.

Figure 2

The figure is a fantasy landscape image with game controls that is completely unrelated to the scientific claims in the caption regarding camera pose training and data filtering. It lacks all necessary scientific visualization elements such as axes, data plots, or legends.

Figure 3

The figure demonstrates visual results of camera control but fails to explicitly map the displayed keyboard inputs to specific camera movements in the caption or via a legend, leaving the viewer to infer the actions.

Figure 4

Figure 4 is a clear and well-structured pipeline diagram that effectively visualizes the minWM framework. The flow from inputs through data, training, and inference stages is logical, and the output examples clearly demonstrate the system's capabilities. The caption accurately describes the figure's content.

Figure 5

The figure is a mismatch for its caption; it shows game interface screenshots rather than the training data or metrics described. Consequently, the visual content fails to provide any evidence for the claim that the model has not acquired camera controllability.

Action items.

- [**fatal**] Figure 1: The rendered image displays a fantasy landscape with video game UI overlays (WASD keys) rather than the scientific data (e.g., loss curves, sample comparisons) described in the caption regarding 'Training with very small batch sizes' and 'Wan2.1-based model' failure.
- [**fatal**] Figure 2: The rendered image displays a fantasy landscape with game UI overlays (WASD keys) rather than scientific data, charts, or training curves. It fails to visually support the caption's claim about 'Training with estimated camera poses' or 'SpatialVid data'.
- [**science**] Figure 2: The image contains no axes, units, legends, or error bars, making it impossible to evaluate the 'reliable camera-controllable generation' or 'perception-estimated camera poses' mentioned in the caption.
- [**writing**] Figure 3: The figure displays a grid of images with overlaid keyboard controls (WASD, arrows) but lacks a legend or explicit labels defining what specific camera action (e.g., pan, tilt, zoom) corresponds to each column or row.
- [**writing**] Figure 3: The caption states the model supports generation under 'different camera actions' but does not describe the specific actions shown in the grid, making it difficult to verify the claim without guessing based on the icons.
- [**fatal**] Figure 5: The rendered image displays three identical screenshots of a desert landscape with game UI overlays (WASD/arrow keys), which does not visually support the caption's claim about 'early-stage training' or 'camera controllability' failure. There are no axes, data plots, or comparative metrics shown to substantiate the scientific claim.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits a high density of specialized acronyms and domain-specific shorthand that hinders accessibility for non-specialist readers, particularly in the Abstract and Method sections.

In the **Abstract**, the term **MMDiT** is used to describe the HY1.5 architecture without expansion. Similarly, **DMD** is introduced as a core component of the pipeline without defining it as "Distribution Matching Distillation." These acronyms are critical to the paper's contribution but are presented as if they are common knowledge, which excludes readers from adjacent fields.

In **Section 3.1 (Camera-Controllable Training)**, the text references the **GTA** form for attention injection without defining the acronym. It also uses **SE(3)** to denote the extrinsic transformation group; while standard in robotics, a brief parenthetical explanation (e.g., "the group of 3D rigid body transformations") would aid general readers.

In **Section 3.2 (AR Diffusion Distillation)**, the term **PF-ODE** is used to describe the trajectories generated by the AR diffusion model. This should be expanded to "Probability Flow Ordinary Differential Equation" at first use. Additionally, **EMA** is used to describe the teacher model update mechanism without expansion.

Finally, the term **Causal Forcing++** is used as a proper noun throughout. While the paper cites the source, the "++" notation is informal. It would be clearer to refer to it as "Causal Forcing++ (an extension of Causal Forcing)" or similar upon first mention to clarify its relationship to the base method.

These issues are purely stylistic and do not reflect on the scientific validity of the work, but they significantly increase the barrier to entry for a broader audience, which is counter-productive for a paper claiming to provide an "open-source framework" and "reproducible recipe."

Action items.

- [**writing**] Define 'MMDiT' at first use in the Abstract and Introduction. While common in diffusion literature, it is an acronym that excludes non-specialist readers without a brief expansion (e.g., 'Mixture-of-Experts Multi-Modal Diffusion Transformer' or similar).
- [**writing**] Define 'DMD' (Distribution Matching Distillation) at first use in the Abstract and Section 3. The term is used as a proper noun without expansion, assuming prior knowledge of the specific distillation technique.
- [**writing**] Define 'PF-ODE' (Probability Flow Ordinary Differential Equation) at first use in Section 3.2. The acronym is used without definition, which may confuse readers unfamiliar with the specific terminology of score-based generative modeling.
- [**writing**] Define 'GTA' (Global Token Attention or similar) in Section 3.1 where the PRoPE injection is described. The text states 'in the GTA form' without explaining what GTA stands for or how it differs from standard attention mechanisms.
- [**writing**] Define 'SE(3)' in Section 3.1. While standard in robotics, explicitly stating 'the special Euclidean group of 3D rigid body transformations' upon first mention improves accessibility for general ML readers.
- [**writing**] Define 'EMA' (Exponential Moving Average) in Section 3.2 when describing the teacher model update. This is a standard optimization term but should be spelled out for

clarity in a framework paper.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent pipeline for converting bidirectional diffusion models into autoregressive world models. However, there are logical gaps in the justification of the distillation objective and the interpretation of latency metrics.

First, in Section 3.2 (Stage 3: asymmetric DMD), the authors state the goal is to align the few-step AR model with the “high-quality distribution of the bidirectional teacher.” Yet, Equation 4 defines the objective as minimizing the KL divergence between the student’s distribution $p_{\theta,t}$ and $p_{\text{data},t}$, estimated by a frozen model s_{real}. If s_{real} is initialized from the bidirectional teacher, it approximates the teacher’s distribution, not the true data distribution. The text conflates “data distribution” with “teacher distribution” without clarifying if the teacher is assumed to be a perfect proxy for data. If the teacher is imperfect, optimizing against it directly (as implied by the text) contradicts the standard DMD formulation which targets the data distribution. This ambiguity weakens the logical support for the claimed “asymmetric” advantage.

Second, the latency analysis in Section 4.2 and Table 1 relies on a definition of “first-frame latency” that creates a logical asymmetry in comparison. The bidirectional baseline latency (e.g., 771s for HY1.5) represents the time to generate the *entire* 77-frame video. The AR baseline latency (e.g., 3.4s) represents the time to generate only the *first* frame (or chunk). While the text acknowledges this difference, presenting a 223x speedup as a measure of “real-time interactive” capability is logically misleading. A fair comparison for interactivity would be the time to generate the first frame of the bidirectional model (which might be near zero if streaming is supported, or the full time if not) versus the AR model’s first frame. The current metric compares “total work” vs. “partial work,” inflating the perceived speedup.

Finally, the ablation on training data (Section 4.3) posits that SpatialVid fails due to noisy estimated poses, while WorldPlay-generated data succeeds. The logical leap here is assuming that WorldPlay’s generated videos provide “effectively ground-truth trajectories.” WorldPlay generates videos *conditioned* on trajectories; it does not measure them. If the WorldPlay model has its own biases or errors in following the trajectory, the “ground truth” is just as synthetic as the SpatialVid estimates. The paper does not logically justify why a generative model’s adherence to a prompt is superior to a perception model’s estimation of a real video, other than an implicit assumption that the generative model is more consistent. This requires a clearer logical argument or empirical validation of the trajectory fidelity.

Action items.

- [**science**] In Sec 3.2 (Stage 3), Eq 4 minimizes KL to p_data, but text claims alignment with the ‘bidirectional teacher’. Clarify if s_real approximates p_data or the teacher’s distribution to resolve the logical conflict in the training signal source.

- **[writing]** Table 1 compares total bidirectional generation time against AR first-frame time. This 'total vs. partial' comparison logically inflates the speedup metric. Clarify if the baseline should be bidirectional first-frame latency for a fair interactivity comparison.
- **[science]** Sec 4.3 claims WorldPlay-generated trajectories are 'effectively ground-truth' while SpatialVid estimates are noisy. Logically justify why a generative model's conditioned output is superior to perception estimates, as both are synthetic/non-physical.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that extend beyond the immediate evidence provided in the text and tables, specifically regarding the definitions of "real-time," the generality of the framework, and the quantitative basis for ablation conclusions.

First, the title and abstract repeatedly characterize minWM as a framework for "Real-Time Interactive Video World Models." However, the empirical results in Table 1 (Section 4.1) report a first-frame latency of 1.137 seconds for the Wan2.1 model and 3.446 seconds for the HY1.5 model. In the context of interactive systems, "real-time" typically implies latencies below 100ms (or at least below the frame interval of 33ms for 30fps) to ensure responsiveness. A delay of over one second creates a noticeable lag that contradicts the standard definition of real-time interaction. While the authors correctly note this is a massive speedup over the bidirectional baseline, extrapolating this to "real-time" is an over-claim. The terminology should be adjusted to "low-latency" or the specific use-case where >1s latency is acceptable must be explicitly defined and justified.

Second, the authors claim the framework is "architecture-extensible" and "modular" (Abstract, Introduction), suggesting it can be easily applied to various video backbones. The evidence provided, however, is limited to two specific instantiations: Wan2.1 (cross-attention) and HY1.5 (MMDiT). There is no discussion or experimental validation regarding the framework's application to other common architectures (e.g., standard U-Nets or different transformer variants) that might require non-trivial modifications to the PRoPE injection or distillation stages. Claiming broad extensibility based on only two successful cases is an over-extrapolation. The text should qualify this claim to reflect that it has been demonstrated on these specific architectures, or provide a theoretical argument for why it generalizes without empirical proof.

Finally, the ablation study on "Minimal batch size" (Section 4.3, Figure 4) concludes that a batch size of 16 is necessary for "high controllability," while batch size 8 is "unstable." This conclusion relies entirely on visual inspection of the generated video samples in Figure 4. The paper lacks a quantitative metric (such as camera pose error, trajectory alignment score, or a user study) to objectively distinguish between "unstable" and "high" controllability. Without such metrics, the specific threshold of 16 is a subjective over-claim. The authors should either provide quantitative data supporting this specific threshold or soften the language to reflect that the finding is based on qualitative observation.

Action items.

- [**writing**] The claim of 'real-time' interaction in the title and abstract is not fully supported by the reported latencies. Table 1 shows a first-frame latency of 1.137s for Wan2.1 and 3.446s for HY1.5. While this is a significant speedup, a >1s delay contradicts the standard definition of 'real-time' (typically <100ms or <33ms for 30fps) in interactive systems. The text should temper this claim to 'low-latency' or explicitly define the specific interactive scenario where >1s latency is acceptable.
- [**writing**] The paper claims the framework is 'architecture-extensible' and 'modular' (Abstract, Intro), yet the experimental validation is limited to only two specific backbones (Wan2.1 and HY1.5). There is no evidence provided that the pipeline works on other architectures (e.g., U-Net based models or different MMDiT variants) without significant re-engineering. The claim of general extensibility is an over-extrapolation from the current limited instantiation.
- [**science**] The ablation on 'minimal batch size' (Sec 4.3) claims that batch size 16 is required for 'high controllability' based on visual inspection of Figure 4. However, the paper does not provide a quantitative metric (e.g., pose error, trajectory alignment score) to substantiate the qualitative jump from 'unstable' (BS=8) to 'high controllability' (BS=16). Without quantitative evidence, the specific threshold of 16 is an over-claim based on subjective visual assessment.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents minWM, a framework for converting video diffusion models into interactive world models. From a safety and ethics perspective, the paper is largely descriptive of a technical pipeline but lacks necessary disclosures regarding the potential for misuse and data provenance.

First, the paper highlights the generation of synthetic training data using WorldPlay on images from OpenVid (Section 4.3, "Training data"). OpenVid is a large-scale dataset scraped from the internet. The authors do not specify any filtering mechanisms applied to these source images to remove harmful, illegal, or copyrighted content before using them to train the video generation model. Without explicit confirmation of data curation and safety filtering, there is a risk that the resulting world models could inadvertently learn and reproduce biases or harmful content present in the raw internet data.

Second, the core contribution is a “real-time interactive” video generation system (Abstract, Introduction). Such technology has significant dual-use potential, including the creation of realistic deepfakes, disinformation campaigns, or non-consensual synthetic media. The manuscript currently treats the technology purely as a scientific advancement without addressing these societal risks. Standard practice for such frameworks requires a dedicated section on “Ethical Considerations” or “Broader Impacts” that discusses potential misuse, the authors’ stance on responsible use, and any technical safeguards (e.g., watermarking, usage policies) implemented or recommended for the open-source release.

Finally, while the authors mention using DL3DV for 3D reconstruction to create ground-truth camera trajectories (Section 4.3), they do not explicitly confirm the licensing and consent status of the source videos used for reconstruction. Given the sensitivity of 3D reconstruction and generative training, a brief statement confirming that the source data was used in accordance with its license and privacy guidelines would be prudent.

The paper should be revised to include a discussion on data safety filtering, potential misuse scenarios, and responsible release guidelines before it can be considered fully compliant with safety and ethics standards.

Action items.

- [**writing**] The paper describes using ‘WorldPlay’ to generate synthetic training videos from OpenVid images (Sec 4.3). As this involves generating realistic video content from potentially uncurated internet images, the authors must explicitly state the data filtering protocols used to prevent the model from learning from or reproducing harmful, biased, or copyrighted content present in the source images.
- [**writing**] The framework enables ‘real-time interactive’ video generation with camera control (Abstract, Sec 1). This capability carries dual-use risks for generating deepfakes or disinformation. The manuscript currently lacks a dedicated ‘Ethical Considerations’ or ‘Limitations’ section discussing potential misuse, mitigation strategies (e.g., watermarking), and responsible release guidelines for the open-source code.
- [**writing**] The ablation study in Sec 4.3 notes that models trained on ‘SpatialVid’ (perception-estimated poses) failed, while those trained on ‘DL3DV’ (reconstructed scenes) succeeded. The authors should clarify the provenance and consent status of the DL3DV dataset, specifically whether the 3D reconstructions were performed on data with appropriate usage rights for training generative models.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The manuscript presents a compelling engineering framework for converting bidirectional video diffusion models into real-time autoregressive world models. However, from a strict scientific evidence perspective, the validation of the core claims relies heavily on qualitative visual inspection and single-point measurements, which introduces significant risks of bias and overfitting to specific hyperparameters.

First, the ablation studies in Section 4.3 (lines 230-260) regarding training steps and batch sizes are the primary evidence for the “practical guidance” contribution. The authors assert that models are “completely uncontrollable” at 2k steps or “fail to learn” at batch sizes <4 based solely on visual examples in Figures `ablation-steps` and `ablation-bs`. Without quantitative metrics—such as the Mean Squared Error (MSE) between the generated camera trajectory and the ground truth, or a perceptual alignment score—these claims are subjective. It is impossible to verify if the “uncontrollable” models are merely noisy or truly failing to condition, or if the “successful” models at 8k steps are simply overfitting to the specific training seeds shown. The lack of statistical replication (e.g., results averaged over 3-5 random seeds with error bars) makes it difficult to distinguish between a robust training dynamic and a lucky initialization.

Second, the latency results in Table 1 (lines 185-198) are presented with high precision (e.g., 1.137s, 236.64x speedup) but lack any measure of uncertainty. Latency in deep learning inference is highly variable due to system noise, memory fragmentation, and GPU scheduling. Reporting a single measurement without standard deviation or a confidence interval undermines the scientific rigor of the performance claim. A robust evaluation would require reporting the mean and standard deviation over a large number of inference runs (e.g., n=100) to establish the stability of the speedup.

Finally, while the instantiation on Wan2.1 and HY1.5 demonstrates the framework’s utility, the claim of being “architecture-general” (Abstract, lines 12-13) is only weakly supported by two data points. While not a fatal flaw, the evidence would be significantly stronger if the authors acknowledged the limited scope of the architectural validation or provided a brief discussion on why these two specific architectures (cross-attention vs. MMDiT) are sufficient to prove generalization.

To elevate the scientific evidence, the authors should replace qualitative assertions with quantitative metrics for controllability, report statistical variance for latency measurements, and clarify the scope of their architectural generalization claims.

Action items.

- [**science**] The ablation studies on training steps (Sec 4.3, Fig. ablation-steps) and batch size (Sec 4.3, Fig. ablation-bs) rely entirely on qualitative visual inspection of generated frames. To support the claim of ‘strong controllability’ or ‘failure to learn,’ quantitative metrics (e.g., camera pose error RMSE, trajectory alignment scores) or statistical significance tests across multiple seeds are required to rule out cherry-picking.
- [**science**] The latency claims in Table 1 (Sec 4.1) report single-run measurements on a single A800 GPU without reporting variance, standard deviation, or confidence intervals. Given the stochastic nature of GPU scheduling and memory allocation, a single measurement is insufficient to substantiate the precise speedup factors (e.g., 236.64x) claimed.
- [**science**] The paper claims the framework is ‘architecture-general’ by instantiating it on

Wan2.1 and HY1.5 (Sec 1, Sec 4). However, the experimental evidence is limited to these two specific models. To robustly support the generalization claim, the authors should either include a third distinct architecture or explicitly qualify the claim to the tested architectures.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a framework for converting video diffusion models into interactive world models. From a statistical analysis perspective, the primary concern is the lack of quantitative rigor in the ablation studies and performance reporting.

In Section 4.3 (Ablation Studies), the authors evaluate the effects of training steps (Fig. 3) and batch size (Fig. 4) on camera controllability. The conclusions drawn—such as the model being "completely uncontrollable" at 1-2k steps or "unstable" at batch size 8—are based entirely on qualitative visual inspection of generated video frames. Without quantitative metrics (e.g., Mean Squared Error between predicted and ground-truth camera trajectories, or a specific controllability score) and statistical measures of variance (e.g., results averaged over multiple random seeds with standard deviations), these claims are subjective and difficult to reproduce. The assertion that a batch size of 16 is the "minimum" for success requires a statistical definition of "success" and a failure rate analysis across seeds.

Furthermore, Table 1 reports first-frame latency as single point estimates (e.g., 3.446s). In systems research, latency is a stochastic variable influenced by GPU scheduling, memory bandwidth, and other factors. Reporting only the mean without standard deviation or confidence intervals makes it impossible to assess the reliability of the claimed $223.75\times$ speedup. A proper statistical treatment would involve running the inference multiple times (e.g., $N \geq 10$) and reporting the mean $\pm$ standard deviation to demonstrate that the observed speedup is statistically significant and not an artifact of a single favorable run.

Finally, the comparison between different data sources (SpatialVid vs. DL3DV vs. WorldPlay) in Section 4.3 relies on visual evidence. To substantiate the claim that "ground-truth camera poses are crucial," the authors should provide a quantitative comparison of the resulting model performance on a held-out test set using a standardized metric, rather than relying on qualitative descriptions of the generated videos.

Action items.

- [**science**] The ablation studies on training steps (Sec 4.3, Fig. 3) and batch size (Sec 4.3, Fig. 4) rely entirely on qualitative visual inspection. To support the claim of 'strong controllability' or 'instability,' report quantitative metrics (e.g., camera pose error RMSE, trajectory alignment scores) with confidence intervals or standard deviations across multiple seeds.

- [**science**] Latency results in Table 1 are reported as single point estimates without variance. Given the stochastic nature of GPU inference and potential system noise, report mean latency and standard deviation over multiple runs to validate the claimed speedup factors.
- [**science**] The claim that batch size < 4 'often fails' (Sec 4.3) lacks statistical definition. Specify the failure rate (e.g., 0/5 seeds failed) and the criteria used to define 'failure' to ensure reproducibility of the threshold.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a clear and well-structured narrative regarding the minWM framework. The abstract effectively summarizes the problem, the proposed solution, and the key contributions. The introduction successfully motivates the need for a unified pipeline, and the logical flow from problem statement to method and results is generally strong. The writing style is professional and appropriate for a technical audience.

However, there are a few specific instances where the writing quality and formatting detract from the overall readability. In Section 3 (Method), the first sentence of the main paragraph contains a missing space after the period ("...video generator.The pipeline..."), which is a basic typographical error. Additionally, the enumeration of the three distillation stages in Section 3.2 is visually cluttered due to the excessive use of bold and italic formatting within the running text. This makes the sentence harder to parse than necessary.

Furthermore, in Section 4.2, the paragraph heading "Few-step AR models substantially reduce the first-frame latency." appears to be merged with the subsequent text in the source code, leading to a formatting issue where the text runs into the heading. This disrupts the visual hierarchy and readability of the results section. While these issues are minor and do not obscure the scientific content, correcting them would significantly improve the polish and professional presentation of the paper. The rest of the manuscript is free of significant grammatical errors or flow issues.

Action items.

- [**writing**] In Section 3 (Method), the first sentence of the main paragraph lacks a space after the period: '...video generator.The pipeline...'. This is a clear typographical error that disrupts readability and should be corrected.
- [**writing**] In Section 3.2, the enumeration of the three distillation stages is formatted awkwardly within the text: '...three stages: mph{ extbf{(1) Stage 1... (3) Stage 3: asymmetric DMD.}}'. The use of bolding and italics for the list items within a running sentence is visually cluttered. Consider simplifying to '...three stages: (1) AR diffusion training; (2) causal ODE or causal CD initialization; and (3) asymmetric DMD.'

- [**writing**] In Section 4.2 (Results), the paragraph heading 'Few-step AR models substantially reduce the first-frame latency.' ends with a period, but the subsequent text begins immediately on the same line without a line break or proper paragraph separation in the source, causing a formatting glitch in the compiled PDF where the text runs into the heading. Ensure proper paragraph breaks are used.